

## SUPERIOR SHOALS, NY, USA

WMO: 994580

Lat: 44.4645N

Lon: 75.7936W

Elev: 77

StdP: 100.40

Time Zone: -5.00 (NAE)

Period: 94-09

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.6      | -17.8 | -27.4                         | 0.3 | -20.1 | -24.0 | 0.4 | -16.9 | 21.0                  | 3.7  | 18.0 | -0.1 | 3.8                   | 40   | 0.630 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.4                    | 27.4            | 21.4 | 26.3 | 20.8 | 25.2 | 20.2 | 23.0                | 25.5 | 22.3 | 24.7 | 21.6 | 24.0 | 5.4                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.3                            | 17.1 | 24.3 | 21.5 | 16.3 | 23.9 | 20.7 | 15.5 | 23.4 | 68.6          | 25.6 | 66.0 | 24.9 | 63.3 | 23.9 | 26.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 17.3          | 14.3 | 11.7 | DB                         | -23.0 | 29.5               | 3.9 | 1.2                                                | -25.8 | 30.4       | -28.1 | 31.1       | -30.3 | 31.7       | -33.2 | 32.6 | (4) |
| (5)               |      |      | WB                         | -23.2 | 24.5               | 4.2 | 1.2                                                | -26.2 | 25.3       | -28.7 | 26.0       | -31.1 | 26.7       | -34.2 | 27.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.1         | -6.2  | -5.3  | -0.4  | 6.3   | 12.7  | 18.2  | 20.9  | 20.6  | 16.8  | 10.1  | 4.3   | -2.2  | (6)  |  |
|                                                                        | DBStd       | 10.57       | 6.86  | 5.64  | 5.35  | 4.45  | 3.37  | 3.10  | 2.38  | 2.59  | 3.57  | 4.22  | 4.70  | 5.48  | (7)  |  |
|                                                                        | HDD10.0     | 2002        | 502   | 428   | 323   | 127   | 13    | 0     | 0     | 0     | 1     | 52    | 178   | 378   | (8)  |  |
|                                                                        | HDD18.3     | 3976        | 761   | 662   | 580   | 361   | 178   | 39    | 6     | 9     | 69    | 256   | 420   | 636   | (9)  |  |
|                                                                        | CDD10.0     | 1294        | 0     | 0     | 1     | 16    | 95    | 247   | 337   | 328   | 206   | 56    | 8     | 1     | (10) |  |
|                                                                        | CDD18.3     | 225         | 0     | 0     | 0     | 0     | 2     | 36    | 84    | 78    | 23    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 768         | 0     | 0     | 0     | 0     | 9     | 117   | 284   | 281   | 77    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 63          | 0     | 0     | 0     | 0     | 0     | 11    | 20    | 24    | 7     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 5.2         | 5.7   | 5.8   | 5.8   | 5.6   | 4.9   | 4.2   | 4.5   | 4.0   | 4.8   | 5.5   | 5.8   | 5.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 982         | 74    | 61    | 65    | 79    | 83    | 82    | 92    | 79    | 92    | 104   | 92    | 84    | (15) |  |
|                                                                        | PrecMax     | 1226        | 154   | 107   | 121   | 144   | 146   | 175   | 198   | 118   | 173   | 182   | 144   | 143   | (16) |  |
|                                                                        | PrecMin     | 782         | 26    | 25    | 28    | 10    | 33    | 18    | 33    | 24    | 49    | 19    | 42    | 37    | (17) |  |
|                                                                        | PrecStd     | 106         | 32    | 18    | 25    | 36    | 34    | 39    | 37    | 31    | 31    | 46    | 26    | 29    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.9  | 9.1   | 16.1  | 21.0  | 24.7  | 28.1  | 28.4  | 28.8  | 27.7  | 22.8  | 17.3  | 13.0  | (19) |  |
|                                                                        |             | MCWB        | 10.8  | 6.3   | 10.5  | 13.1  | 17.6  | 20.7  | 22.5  | 21.8  | 20.5  | 17.1  | 12.9  | 10.8  | (20) |  |
|                                                                        | 2%          | DB          | 7.7   | 6.6   | 11.9  | 18.1  | 21.9  | 26.1  | 27.1  | 27.3  | 25.5  | 20.4  | 15.4  | 9.5   | (21) |  |
|                                                                        |             | MCWB        | 6.6   | 4.7   | 7.3   | 11.5  | 15.7  | 20.3  | 21.8  | 21.3  | 19.7  | 16.4  | 12.3  | 8.0   | (22) |  |
|                                                                        | 5%          | DB          | 5.1   | 4.4   | 9.2   | 15.5  | 20.1  | 24.7  | 26.1  | 26.1  | 23.7  | 18.5  | 13.5  | 7.3   | (23) |  |
|                                                                        |             | MCWB        | 3.7   | 2.6   | 6.1   | 10.4  | 14.3  | 19.5  | 20.9  | 20.7  | 19.2  | 15.1  | 10.9  | 5.8   | (24) |  |
|                                                                        | 10%         | DB          | 2.8   | 2.5   | 6.8   | 12.9  | 18.3  | 23.1  | 25.0  | 25.0  | 22.2  | 16.6  | 11.0  | 5.1   | (25) |  |
|                                                                        |             | MCWB        | 1.6   | 0.9   | 4.1   | 8.6   | 13.8  | 18.9  | 20.4  | 20.4  | 18.3  | 13.5  | 8.7   | 3.6   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.8  | 7.3   | 11.8  | 15.5  | 20.2  | 22.8  | 24.1  | 24.5  | 22.4  | 19.0  | 14.8  | 11.3  | (27) |  |
|                                                                        |             | MCDB        | 12.4  | 8.1   | 15.2  | 18.1  | 22.5  | 25.7  | 26.6  | 26.6  | 25.6  | 21.6  | 16.3  | 12.9  | (28) |  |
|                                                                        | 2%          | WB          | 6.8   | 5.0   | 8.6   | 13.2  | 17.7  | 21.7  | 23.2  | 22.9  | 21.0  | 16.8  | 13.0  | 8.2   | (29) |  |
|                                                                        |             | MCDB        | 7.8   | 6.0   | 10.4  | 16.3  | 20.6  | 24.4  | 25.6  | 25.3  | 23.6  | 19.3  | 14.7  | 9.2   | (30) |  |
|                                                                        | 5%          | WB          | 3.8   | 2.8   | 6.5   | 11.1  | 16.2  | 20.7  | 22.5  | 22.3  | 20.2  | 15.4  | 11.2  | 6.0   | (31) |  |
|                                                                        |             | MCDB        | 5.0   | 4.4   | 9.0   | 14.3  | 18.5  | 23.2  | 24.9  | 24.5  | 22.8  | 17.8  | 13.3  | 7.3   | (32) |  |
|                                                                        | 10%         | WB          | 1.8   | 0.9   | 4.5   | 9.0   | 14.8  | 19.8  | 21.7  | 21.6  | 19.2  | 14.0  | 9.2   | 3.9   | (33) |  |
|                                                                        |             | MCDB        | 2.8   | 2.0   | 6.7   | 12.8  | 17.2  | 22.5  | 24.0  | 23.8  | 21.8  | 16.4  | 10.9  | 5.1   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.5   | 8.2   | 7.5   | 7.8   | 7.9   | 7.2   | 6.4   | 6.7   | 7.3   | 6.4   | 5.8   | 6.3   | (35) |  |
|                                                                        |             | MCDBR       | 8.7   | 9.2   | 11.1  | 11.8  | 10.8  | 8.8   | 7.5   | 7.5   | 8.7   | 8.7   | 9.5   | 8.1   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 8.2   | 8.1   | 7.9   | 6.9   | 5.7   | 4.5   | 4.3   | 4.1   | 4.9   | 6.0   | 7.8   | 7.2   | (37) |  |
|                                                                        |             | MCWBR       | 8.8   | 9.2   | 10.9  | 10.5  | 9.3   | 7.4   | 6.4   | 6.2   | 7.5   | 8.0   | 9.0   | 8.2   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.290 | 0.295 | 0.324 | 0.354 | 0.383 | 0.417 | 0.429 | 0.418 | 0.368 | 0.349 | 0.310 | 0.285 | (40) |  |
|                                                                        |             | taud        | 2.373 | 2.359 | 2.417 | 2.395 | 2.357 | 2.277 | 2.249 | 2.281 | 2.408 | 2.456 | 2.508 | 2.452 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 831   | 895   | 914   | 908   | 887   | 854   | 838   | 834   | 854   | 823   | 808   | 805   | (42) |  |
|                                                                        |             | Edh at Noon | 76    | 92    | 101   | 112   | 120   | 131   | 133   | 125   | 101   | 85    | 68    | 64    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.35  | 2.23  | 3.56  | 4.41  | 5.28  | 5.65  | 5.78  | 5.09  | 4.05  | 2.41  | 1.51  | 1.04  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.13  | 0.23  | 0.29  | 0.54  | 0.55  | 0.45  | 0.50  | 0.23  | 0.34  | 0.23  | 0.16  | 0.12  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (31 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page